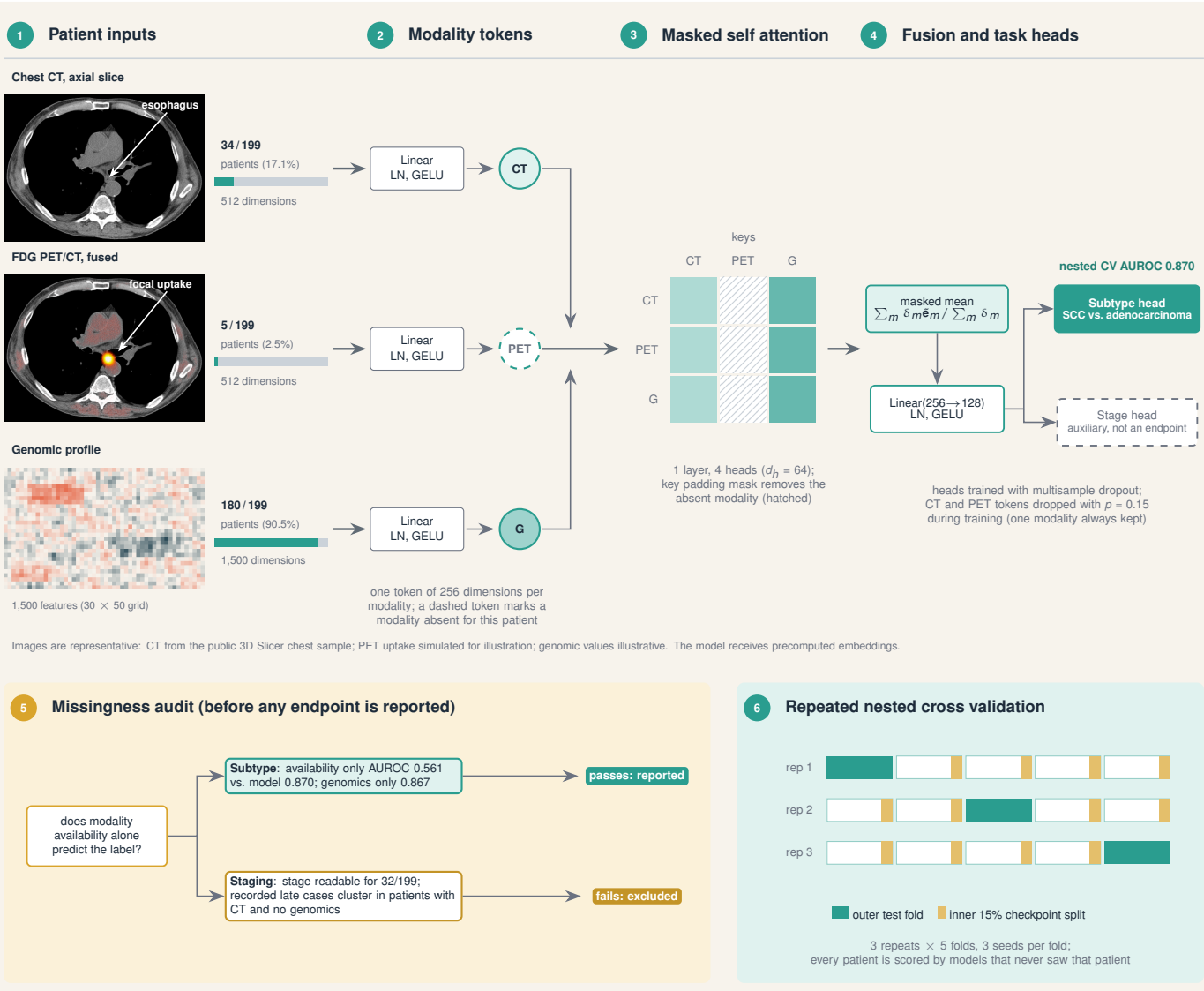

1 Patient inputs
2 Modality tokens
3 Masked self attention
4 Fusion and task heads
Chest CT, axial slice
esophagus
FDG PET/CT, fused
focal uptake
Genomic profile
1,500 features (30 × 50 grid)
34 / 199
patients (17.1%)
512 dimensions
5 / 199
patients (2.5%)
512 dimensions
180 / 199
patients (90.5%)
1,500 dimensions
Linear
LN, GELU
Linear
LN, GELU
Linear
LN, GELU
CT
PET
G
one token of 256 dimensions per modality; a dashed token marks a modality absent for this patient
keys
CT
PET
G
CT
PET
G
1 layer, 4 heads (d_h = 64); key padding mask removes the absent modality (hatched)
masked mean
$\sum_m \delta_m \tilde{\mathbf{e}}_m / \sum_m \delta_m$
Linear(256→128)
LN, GELU
nested CV AUROC 0.870
Subtype head
SCC vs. adenocarcinoma
Stage head
auxiliary, not an endpoint
heads trained with multisample dropout; CT and PET tokens dropped with p = 0.15 during training (one modality always kept)
Images are representative: CT from the public 3D Slicer chest sample; PET uptake simulated for illustration; genomic values illustrative. The model receives precomputed embeddings.
5 Missingness audit (before any endpoint is reported)
does modality availability alone predict the label?
Subtype: availability only AUROC 0.561 vs. model 0.870; genomics only 0.867
passes: reported
Staging: stage readable for 32/199; recorded late cases cluster in patients with CT and no genomics
fails: excluded
6 Repeated nested cross validation
rep 1
rep 2
rep 3
outer test fold
inner 15% checkpoint split
3 repeats × 5 folds, 3 seeds per fold; every patient is scored by models that never saw that patient